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A Digital Twin With Material Characterisation and LLM as High-Level Orchestrator via Few-Shot In-Context Learning for Closed-Loop Control in Robotic Adaptive 3DCP

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ABSTRACT

The industrialisation of 3D concrete printing (3DCP) is impeded by the lack of universal standards, a situation exacerbated by the process's sensitivity to multiple interdependent parameters and environmental conditions, which can introduce structural defects and performance inconsistencies. Starting from the definition of 3DCP as a cyberphysical production system (CPPS), this exploratory study proposes adopting a hybrid standardisation method that optimises structural performance, or concrete grade. By following the DSR/ASTM V&V research framework and based on the preliminary characterisation of materials, a computational digital twin is implemented, including a compressive strength predictive model connected to the simulation of the adaptive 3DCP process. Accounting for the anisotropic behaviour of 3D-printed concrete (3DPC), the predictive model determines the 3DPC grade based on preliminary material characterisation results and considering different process factors that determine the mechanical performance: By optimising process parameters, the model maximises parallel compression strength, prioritising safety. Uncertainty propagation is embedded in the computational model, enabling statistical validation through extensive virtual testing. The optimised configuration of process parameters will serve as a benchmark for the adaptive 3DCP process, in which a pretrained large language model acts as a high-level orchestrator for closed-loop control via few-shot in-context learning. The proposed architecture relies on the performance predictive model, sensor readings, and an LLM that interprets the process status and suggests corrective actions. The primary novel contribution of this work is a computational performance predictive model with uncertainty propagation; the adaptive 3DCP is described as completing the digital twin definition and demonstrating its use in a production setting. Starting from preliminary materials characterisation and considering the specificities of our robotic cell and environmental conditions, the performance predictive model reduced the variance in parallel structural strength by 25%, ensuring a 90% confidence level in achieving the M20 grade, with a significant reduction in anisotropic behaviour. Additionally, while maximising 3DPC performance, the digital twin enables benchmarking of production parameters across any production setting, making it a valid tool for standardising the technology. This research proposes a critical transition from deterministic to flexible standardisation, and from training-for-control to a more versatile prompting-for-control AI methodology, offering a viable, data-efficient pathway towards standardised, reliable, and industrial-scale 3DCP.

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1 | Introduction

The construction industry is transitioning from fragmented, craft-based methodologies to integrated cyberphysical production systems (CPPS) [1]. This transformation is driven by digital fabrication, which serves as a critical physical instantiation of the CPPS framework, providing an essential link between digital design data and physical output [2]. The core of this paradigm is the closed-loop cyberphysical process, where robotic systems, governed by digital instructions, are enhanced by integrated sensors that provide real-time feedback, enabling instantaneous in-process corrections [3, 4]. This data exchange enables fabrication to transition from a static operation to an adaptive production process [5]. CPPS are characterised by high complexity, driven by many interconnected parameters that ultimately affect the quality of the final product; traditional automation based on preset rules fails under complex, nonlinear, and unpredictable conditions. In these cases, integrating artificial intelligence (AI) is a compelling solution. The adoption of specialised, 'expert' AI models—convolutional neural networks (CNNs) for perception, long short-term memory networks (LSTMs) for sequence prediction, and reinforcement learning (RL) for control—trained on large, curated, and annotated datasets of defects, has proven to be effective for simple tasks; however, this approach is constrained by hardware requirements and the cost of data acquisition and model training, creating a formidable barrier to entry [6]. Furthermore, any shift in the operational environment or a requirement for a new capability often mandates a complete retraining pipeline with newly collected data: This inherent narrowness limits generalisability across different setups and defect types [7].

A transformative alternative emerges with large language models (LLMs) and multimodal LLMs (mLLMs). These foundation models, built on the transformer architecture, are pretrained on vast, diverse corpora and exhibit remarkable general-purpose reasoning capabilities [8, 9]. The transformer's core mechanism, the *attention model*, enables it to contextually weigh the relevance of all elements in a sequence—whether words in a sentence or patches in an image—allowing for a holistic understanding that mirrors the interconnected nature of complex systems [10].

The profound reasoning and planning capabilities of LLMs and mLLMs are increasingly being leveraged to tackle long-standing challenges in embodied AI. Rather than directly controlling actuators, these models often act as high-level reasoning engines. A prominent application is in task planning, where a high-level natural language instruction is decomposed by an LLM into a sequence of executable subtasks. Furthermore, LLMs have shown promise in code generation, where a natural-language command is translated directly into executable code for robot control [11]. The integration of LLMs enables robots to reason about their physical environment through visual input, a critical step towards generalisable autonomy.

Governed by a considerable number of interdependent design, environmental, and production parameters, and extremely sensitive to any variation in these parameters, 3D concrete printing (3DCP) exemplifies a complex CPPS in the modern construction industry. In 3DCP, a robotic system precisely controls the deposition of material, translating digital geometries into physical structures layer by layer [12], and allowing the sustainable production of bespoke, structurally optimised elements with a high degree of geometric freedom [13]. Beyond 3DCP, robotic fabrication for construction spans a wide spectrum of applications, including automated bricklaying and, more generally, prefabricated component assembly, as well as subtractive machining of building elements, all unified by the principle of direct digital-to-physical translation [14].

The main issue with 3DCP as a complex system is determining standardised processes to systematically obtain the required mechanical performance (grade) of 3D-printed concrete (3DPC) independently of the specific materials and hardware/software setup. The material is, in fact, anisotropic and contingent upon precise control of design, production and environmental parameters [15], with different failure modes and typologies [16]. Traditional automation, based on preset rules, fails under the complex, nonlinear and unpredictable conditions characteristic of both indoor and outdoor fabrication settings [17]. In this context, the principles of high-level planning and multimodal reasoning demonstrated by LLMs in robotics applications share a conceptual synergy with advanced automation, as shown recently with the proposed decentralised, AI-driven workflows for standardising and optimising 3DCP [18–20]: In both contexts, the LLM serves as an intelligent intermediary that translates high-level goals into a structured, executable robotic process.

This paper posits that general-purpose LLMs, conditioned not through training or fine-tuning but through advanced prompt engineering, can act as high-level orchestrators for adaptive 3DCP (A3DCP). The proposed framework utilises an mLLM/LLM pair to perform few-shot in-context reasoning on real-time sensor data, diagnosing process health and suggesting corrective actions. This approach bypasses the need for specialised AI agents, instead leveraging the broad interdisciplinary knowledge and reasoning capabilities of foundational models to tackle the 3DCP CPPS (Figure 1).

Due to health and safety concerns and the inherent risks of robotic fabrication, amplified by AI-driven automatic adjustments, this seminal work was conducted by implementing a digital twin of the A3DCP process. The structure of the digital twin, described in depth in the methodology section, comprises material characterisation, optimisation, uncertainty propagation, statistical analysis and an AI-enabled simulated production process. In the absence of test-bed experimental data, ensuring the credibility of the digital twin is essential; therefore, a rigorous approach was adopted, thoroughly described in the research framework section.

1.1 | Clarification of Terminology

In this work, the terms 'zero-shot' and 'few-shot' refer exclusively to prompt-based in-context learning for LLMs. 'Few-shot' describes the practice of providing a small number of illustrative examples within the model's context window to specify a task, without updating the model's internal weights. This is distinct from the strict machine learning paradigm of 'few-shot learning', which involves meta-training a model to adapt its parameters from a few examples. Our approach leverages the LLM's emergent ability to infer patterns from context [10]. Correspondingly, 'zero-shot reasoning' refers to providing the model with only a task description or instruction, without any examples. The design of these prompts—encompassing both zero-shot

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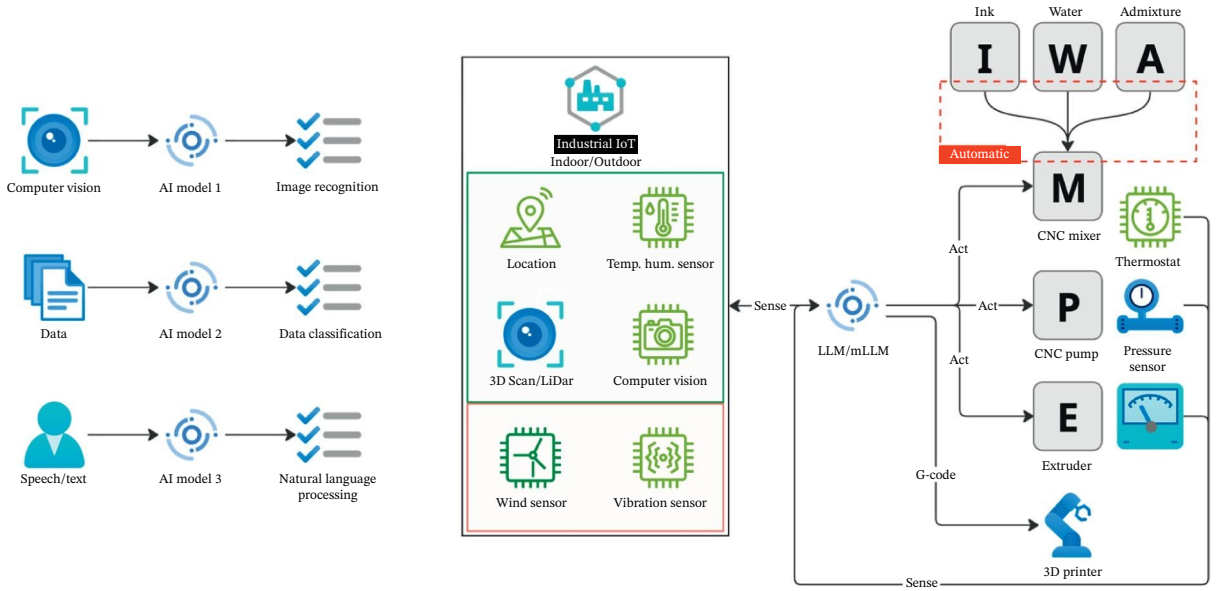


FIGURE 1 | AI architecture based on specialised models (left) and LLM as orchestrator (right).

and few-shot strategies—falls under the broader practice of prompt engineering.

2 | Background Research

The core problem in adopting 3DCP in the construction industry is obtaining the required final grade of the hardened material in accordance with the intended destination of the construct. Traditional cast concrete offers a clear solution based on the concrete mix recipe, where a specific ingredient ratio is sufficient to achieve the desired final grade. With 3DCP, on the other hand, the combination of new engineered materials, materials' fresh and dry rheology, and advanced robotic technologies, makes the standardisation of the process and the deterministic prediction of the final grade challenging.

In this section, we introduce and analyse the elements that define A3DCP, including standardisation models, the definition of CPPS, the characterisation of 3DCP as a CPPS, the differences between specialised AI models and LLMs and the few-shot in-context learning technique enabled by prompt engineering. The literature review demonstrates that the inherent complexity of 3DCP renders traditional automation and specialised AI models insufficient, thereby creating a compelling rationale for adopting general-purpose LLMs as high-level orchestrators for adaptive control.

2.1 | The Hybrid Standardisation Model

The scientific management of industrial production requires a robust approach to standardisation, which can be fundamentally characterised by two distinct philosophical paradigms: the prescriptive and the flexible.

The prescriptive approach is rooted in the principles of scientific management and statistical control [21]. This paradigm seeks to achieve uniformity, interchangeability and repeatability by specifying exact dimensions, values, tolerances and processes, operating on the premise that variation is a primary source of quality defects. Its core objective is to eliminate deviation and ensure perfect conformity to a predefined specification. This is embodied in models such as technical specifications, which provide formal, detailed descriptions of requirements [22], and geometrical product specifications (GPS), a system of international standards that codifies the unambiguous definition of geometrical features like form, position and size on technical drawings [23, 24]. Similarly, the concept of standardised work within lean production systems, which prescribes the precise sequence of tasks, cycle time and in-process inventory for an operator, establishes a controlled baseline that is essential for quality and efficiency [25].

In contrast, the second paradigm is a flexible approach, informed by systems theory and the need for strategic adaptability [26, 27]. This approach establishes goals, performance criteria or interfaces, while granting significant autonomy in the methods used to achieve them. The objective is to ensure interoperability and goal congruence across diverse and dynamic production settings without jeopardising innovation. Key models include performance-based standards, which specify required functional outcomes without mandating the technical solution, thereby fostering innovation in design and materials [28]. Another powerful model is modular design or platform strategy, which standardises key interfaces and core components (the platform) while allowing variation in peripheral modules, enabling mass customisation and product variety [29]. Furthermore, meta-standards, such as ISO 9001, standardise processes for managing an organisation (e.g., quality management systems) rather than prescribing specific products or outcomes, requiring organisations to adapt the principles to their unique context [30]. Table 1 shows a comparative analysis of prescriptive and flexible standardisation features.

Prescriptive standards are typically employed for critical, high-volume or safety-critical components to ensure baseline quality and interchangeability. Flexible standards govern the system and architectural level, enabling product variety, process innovation and resilience to change. This hybrid approach, which leverages the control of prescriptive models and the adaptability of flexible

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TABLE 1 | Comparative analysis of prescriptive and flexible standardisation paradigms.

Feature	Prescriptive standardisation	Flexible standardisation
Philosophical basis	Scientific management, control theory	Systems theory, modularity
Primary focus	Means & methods (the "how")	Ends & interfaces (the "what" and "why")
Core objective	Minimise deviation, ensure conformity	Enable alignment and goal achievement while permitting adaptation
Typical models	Technical specifications, geometrical product specifications, standardised work	Performance-based standards, modular platforms, meta-standards (e.g., ISO 9001)
Optimal application	High-volume, repetitive, safety-critical, and interchangeable components.	Innovative, complex, or variable production environments requiring customisation.
Primary risk	Stifles innovation; processes can be brittle in the face of change.	Potential for misalignment if goals are unclear; requires greater skill to implement effectively.

models, is a hallmark of robust, competitive industrial operations [27].

2.2 | CPPS

The profound transformation of the construction industry is driving a decisive shift away from its traditional reliance on fragmented, craft-based methodologies toward integrated, data-centric frameworks of CPPS, a core manifestation of Industry 4.0 principles in the built environment [1]. To fully comprehend CPPSs, one must first understand their conceptual foundation in cyberphysical systems (CPS). Fundamentally, a CPS is an engineering discipline focused on the deep, organic integration of computation, networking and physical processes, in which embedded computers and networks monitor and control physical entities, with feedback loops that allow physical processes to affect computations and vice versa [31]. The theoretical underpinnings of CPS are deeply rooted in cybernetics, which first formalised the concepts of communication and control in complex, adaptive systems through the fundamental mechanism of feedback loops [32]. In a modern context, CPS are characterised by the tight conjoining and coordination between the system's computational core—the 'cyber' part—and its physical processes, which are monitored and controlled by sensors and actuators through a closed, feedback-driven loop, creating a symbiotic relationship between the digital and physical worlds [33]. A CPPS, therefore, represents the application of this foundational CPS theory specifically to production and manufacturing environments, creating smart, autonomous systems that can dynamically respond to changing conditions in the fabrication lab or construction site. This evolution in construction is critically enabled by digital fabrication, with robotic fabrication serving as the ultimate physical embodiment of the CPPS framework, a robust bridge between digital design data and tangible physical output [2, 34]. The operational core of this advanced paradigm is the implementation of a closed-loop cyber-physical process, a direct realisation of cybernetic feedback, where automated robotic systems, governed by digital instructions, are paired with a dense network of sensors—such as laser scanners, force-torque sensors and computer vision systems—providing a continuous stream of real-time data related to the fabrication process, enabling instantaneous in-process corrections and adjustments that account for real-world variability [35]. The bidirectional data exchange between the computational model and the physical actuator transforms fabrication from a static, predetermined operation into a highly adaptive and responsive production process, capable of accommodating material inconsistencies, geometric deviations and on-site uncertainties [5]. However, deploying a CPPS introduces significant challenges due to its inherent systemic complexity. In fact, when CPPS are characterised by a high degree of complexity due to many interconnected and nonlinear parameters, traditional automation, which relies on rigid, preset rules, becomes inadequate. This fundamental limitation of conventional control strategies highlights the need for more intelligent, self-organising and resilient control paradigms, rooted in CPS theory and a flexible standards approach, to manage the emergent and complex behaviours of robotic construction, ensuring precision and robustness of the processes as well as in the built outcome by providing the production system itself with adaptive capabilities.

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2.3 | 3DCP as a Complex CPPS

The Architecture, Engineering, and Construction (AEC) industry is increasingly embracing digital fabrication technologies, with 3DCP representing an advanced manifestation of this trend. 3DCP offers unprecedented geometric design freedom, enabling the creation of complex, optimised structures that were previously economically or technically unfeasible [36]. This capability is intrinsically linked to significant gains in material efficiency, as the additive, layer-wise deposition process enables material placement only where structurally needed, thereby reducing waste and promoting sustainable construction practices. However, this promise is counterbalanced by the technology's fundamental nature as a complex CPPS, in which performance is not determined by a single material property but by the intricate interplay of multiple parameters. The inherent complexity of 3DCP arises from the interdependencies among the fresh-state rheology of the cementitious material, environmental conditions during printing and curing, and the precise control of the robotic deposition process [37]. Moreover, the rheological properties, which must simultaneously ensure extrudability, shape retention and interlayer adhesion, are highly sensitive to ambient factors such as temperature and humidity. Concurrently, the robotic system's parameters—including printing speed, nozzle geometry and layer deposition strategy—directly influence the structural layered morphology and the resulting mechanical properties. Even the curing environment and conditions can affect the final grade of 3DPC. This interconnectedness of parameters leads directly to anisotropic behaviour and performance variability in the hardened state, a critical challenge for structural reliability. The mechanical performance, particularly in the direction parallel to the layers, is governed by the strength of the interlayer bond, or 'cold joint', which becomes a plane of weakness [38]. Consequently, the ultimate properties of a 3DPC element are not isotropic or perfectly predictable as in cast-in-place concrete but are a direct function of the specific printing process and environmental conditions.

This inherent variability presents a fundamental standardisation problem, as current prescriptive-based codes are not tailored to the performance-based, process-dependent nature of 3DCP as a CPPS. Recent research is now addressing this gap by proposing novel frameworks for standardisation, moving towards decentralised workflows that can reliably link specific printing parameters to final mechanical performance and material grades, which is a critical step for the broader industrialisation of the technology [18, 19].

2.4 | Specialised AI Models

The control of complex CPPS presents formidable challenges that expose the profound inadequacy of traditional computational paradigms. Deterministic automation, in fact, which relies on preprogrammed logic and linear cause-and-effect relationships, is fundamentally unsuited to managing the nonlinear, stochastic and high-dimensional nature of processes like 3DCP. Such processes run under unpredictable conditions in which dry material properties, wet material rheology, environmental factors and hardware dynamics interact in complex ways, leading to emergent failure modes that cannot be individually foreseen or addressed.

In these cases, integrating AI is a compelling solution. The adoption of specialised, expert AI models—CNNs for perception, LSTMs for sequence prediction and toolpath deviations, and RL for closed-loop control—trained on large, curated and annotated datasets of 3DCP defects, has proven to be effective; however, this approach is constrained by the cost of data acquisition and model training, creating a formidable barrier to entry [6]. Furthermore, any shift in the operational environment or additional requirements for new capabilities often necessitates specific retraining using newly collected annotated data; this inherent narrowness limits generalisability across different printing setups and defect types [7].

While representing an advancement over deterministic methods, the adoption of specialised AI creates a fragmented landscape of isolated expert models that lack the holistic understanding, robust reasoning and flexible adaptation required for the integrated structural control and health monitoring of truly autonomous CPPS.

2.5 | Foundational AI Models

A transformative alternative to specialised AI models emerges with LLMs. These foundational models, built on the transformer architecture, are pretrained on vast, diverse datasets and are characterised by remarkable general-purpose reasoning capabilities [8, 9]. The transformer's core mechanism, the attention model, enables it to weigh the relevance of all elements in a sequence—whether words in a sentence or patches in an image—allowing for a contextual understanding that mirrors the interconnected nature of complex systems [10]. The reasoning and planning capabilities of LLMs are increasingly being leveraged to tackle challenges in robotics and embodied AI. Rather than directly controlling actuators, these models often act as high-level reasoning engines. A prominent application in robotics is task planning, where a high-level natural-language instruction can be decomposed by an LLM into a sequence of executable subtasks, accounting for object properties and physical constraints [39]. Furthermore, LLMs have shown promise in robotics code generation, where a natural-language command is translated directly into executable code for robot control [40]. The integration of mLLMs enables robots to reason about their physical environment through visual input, a critical step towards generalisable autonomy.

2.6 | Closed-Loop Control

The pursuit of robust, real-time control for 3DCP constitutes a critical frontier in construction robotics, driven by the imperative to ensure geometric fidelity and the corresponding structural integrity in printed constructs [17]. The process is challenged by the time-dependent rheology of fresh concrete and stochastic material deposition, both of which affect the precise execution of predefined toolpaths and, in turn, the corresponding structural performance [41]. While research has progressed towards sensor-based feedback systems, contemporary closed-loop approaches are typically designed to tackle limited sets of preidentified errors [17]. This specialisation reveals a fundamental lack of generalisability, leaving systems unable to adapt to novel defects or to significant parametric shifts [41]. This limitation highlights an important research gap: the need for a control paradigm capable of zero-shot in-context reasoning, enabling

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a system to interpret and act on unforeseen scenarios without prior explicit training [42, 43]. A conceptual foundation for such a paradigm may be found in recent, transformative advances within artificial intelligence. Specifically, the emergence of large-scale generative pretrained transformer (GPT) models has demonstrated profound capabilities in multimodal understanding and generalisable reasoning, establishing a new technological baseline for adaptive, knowledge-driven systems [42]. The integration of such foundational models into physical control loops represents a nascent but highly promising research frontier [10].

2.6.1 | Reasoning Quality

While the general advantages of LLMs over small or specialised models have been described, adopting LLMs as actuator controllers remains a spearhead research topic, posing challenges that must be addressed before LLM-controlled systems can be deployed in industrial production settings. The ongoing research is promising, and there are many studies on LLM reasoning capabilities across diverse industrial areas [44, 45], demonstrating that LLMs adopting reasoning-optimised architectures (Table 2) implement extended CoT mechanisms with internal verification loops that improve multistep physical reasoning. Given the requirement for multimodal understanding, reasoning-optimised LLMs are the best candidates for 3DCP applications.

2.6.2 | Reproducibility

This property determines LLMs' usability in contexts where safety and compliance are paramount, such as automated construction systems, and is related to the nondeterministic, stochastic responses typical of these AI models. Features like the temperature parameter allow for reducing this behaviour: Low temperature increases determinism and reduces solution diversity, which in the case of control systems is highly desirable. It is important to highlight that near-absolute reproducibility is unachievable by any LLM without an external validation layer.

2.6.3 | Latency

Real-time corrections are a prerequisite for deploying LLMs as controllers in automated processes, especially when standards compliance and safety are involved. The latency of LLMs, typically around 100 milliseconds, poses a significant constraint for structural control applications that require real-time feedback. This delay is prohibitive for systems subject to stringent precision and tolerance specifications or where direct safety implications are present. Conversely, for structural processes characterised by slow dynamics, permissible tolerances, or where primary safety assurances are managed by dedicated systems, a latency of 100 milliseconds may be considered acceptable.

Translating LLMs' potential to the specific and demanding domain of 3DCP remains entirely unexplored [41]. It is this critical gap between the zero-shot in-context reasoning capabilities of advanced AI models and the pressing need for adaptive control in additive construction that the present research seeks to address, proposing a novel methodology to realise a generalised intelligent control framework for 3DCP.

2.7 | Prompt Engineering Techniques

Prompt engineering has emerged as a pivotal discipline for effectively harnessing LLM capabilities in complex, data-driven domains such as structural health monitoring and adaptive control systems. Rather than retraining model parameters, prompt engineering focuses on the strategic design of input queries to guide pretrained models towards domain-specific reasoning and task execution.

Foundational techniques include zero-shot and few-shot in-context learning (ICL), where the LLM is guided towards the expected outputs by providing one or a few examples of input–output correlation within the prompt, thereby adapting its vast knowledge to niche applications without preliminary fine-tuning [42]. This is complemented by the chain-of-thought (CoT) mechanisms [46], which allow the model to articulate its reasoning step-by-step (a feature that has recently become more evident with the 'deep think' option in several GPTs), thereby providing more accurate, logically coherent and verifiable responses. The application of such structured reasoning is critical in domains requiring precise analysis, such as real-time defect detection in additive manufacturing, where understanding the causal chain of a failure is paramount [17]. In multimodal contexts involving visual data from sensors or cameras, image segmentation (IS) is a crucial preprocessing step that enables mLLMs to systematically analyse images by dividing them into 2D segments. This piecewise analysis, as demonstrated in defect detection for 3DCP, has proven to enhance the accuracy of visual reasoning tasks [17, 47, 48]. Building upon CoT, the reason + act (ReAct) framework synergises reasoning with action. ReAct prompts an LLM to alternate verbal reasoning traces with actionable steps: The 'reason' step allows the model to track its plan and internal state, while the 'act' step interacts with external systems to gather new information. This create-plan-act cycle enables the model to handle dynamic environments where

TABLE 2 | Comparative analysis of LLMs' reasoning quality, reproducibility and latency.

Model	Reasoning quality	Reproducibility (%)	Latency (ms)
OpenAI o1-mini	Good—handles 2–3 sensor constraints	92.1	180–420
OpenAI o3-mini	Excellent—solves 5+ constraint optimisation	91.3	210–2300
DeepSeek-R1-Zero	Very good—strong constraint propagation; rarely hallucinates	95.3	110–140
DeepSeek-R1	Excellent—flags uncertainty, does not issue wrong commands	94.8	115–145
Qwen QwQ-32B	Very good—strong physical constraint grounding	93.7	120–180
GLM-Z1-32B-0414	Good—handles 3–4 sensor constraints	94.1	95–130
GLM-4.5V	Very good—strong multimodal reasoning	89.7	150–1800

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preloaded knowledge is insufficient. The ReAct paradigm is a cornerstone for developing autonomous, multistep workflows, such as those proposed for decentralised scientific processes and adaptive manufacturing systems, where an LLM must reason about sensor data and then act by adjusting printing parameters [17, 41] or querying a material database.

The synthesis of these methods enables a form of 'conversational' interaction with AI, where carefully engineered prompts allow models to function as proactive partners in analysing sensor data, diagnosing anomalies and even suggesting corrective actions in CPPS [17, 49]. This evolving paradigm, which moves systems from being merely reactive to becoming suggestive, holds significant promise for creating more intelligent, autonomous and reliable monitoring and control frameworks.

2.8 | Formal Grammars

The industrial automation of processes such as A3DCP increasingly relies on LLMs' interpretative and suggestive capabilities for real-time quality assessment and process adjustment. These models can analyse multimodal sensor data and use natural language to suggest high-level corrective actions. However, a critical challenge lies in the semantic gap between the LLM's high-level output and the deterministic, low-level machine commands required for robotic control and actuation of industrial internet of things (IIoT) equipment [39, 50].

The application of formal grammars presents a practical solution, drawing upon long-established principles in robotics and formal methods [51]. A formal grammar defines a set of syntactic rules that are provided to the AI through prompt engineering and ICL, constraining the LLM's output to a standardised, machine-readable vocabulary and effectively translating high-level process adjustment suggestions—such as 'increase material flow'—into unambiguous, formatted commands [41]. With an eye to trustworthiness and to the eventual industrial adoption of AI-orchestrated fabrication systems, this approach provides a solid base for standardising adaptive control loops, offering a clear, auditable trail from the AI's suggestive analysis to the physical actuator's response.

3 | Research Gaps, Aims and Objectives

From the background research, two main gaps are identified: the lack of a flexible standardisation paradigm for 3DCP, and the need for an effective closed-loop control system for 3DCP capable of assessing, interpreting and controlling the interconnected dimensions of the process. Our general aim is to design a flexible standardisation framework based on a novel computational and AI-enabled architecture with:

- Definition of performance-based material characterisation.
- LLM-orchestrated A3DCP.

This work focuses on the design of a rigorous digital twin, including a scalable performance predictive model for compression strength tailored to materials, technological infrastructure and existing codes and regulations.

To complete the digital twin, an LLM-orchestrated A3DCP process is outlined, providing bidirectional information transfer between the digital twin and the physical model and, at the same time, constituting a first attempt to use reasoning-optimised models instead of specialised AI models for 3DCP control.

3.1 | Research Framework

To ensure the high quality of the digital twin design, simulation and results, we adopted the frameworks illustrated in Table 3.

The uncertainty quantification (UQ) and proof-of-concept (PoC) are explicitly included in the adopted workflow. The overarching structure, core methodology and evaluation, which are fundamental for a digital twin-based study without experimental validation, are based on the engineering-focused digital twin credibility assessment, which blends design science research (DSR) with a rigorous verification & validation (V&V) framework for computational models.

Based on the DSR/V&V research framework, the goal is to ensure that the digital twin is rigorously constructed and constitutes a credible artefact for exploring the research objective, and to define its applicability boundaries. The DSR frames the digital twin as an engineered artefact; the American Society of Mechanics Engineers (ASME) Verification & Validation (V&V) 10-2019, 'Standard for Verification and Validation in Computational Solid Mechanics', provides a rigorous, recognised engineering framework. While focused on solid mechanics, the V&V principles are directly applicable to the structural and material aspects of 3DCP. The DSR/V&V combined framework systematically addresses the predictive credibility of the A3DCP digital twin for supporting specific engineering decisions.

3.2 | Methodology

In compliance with the newly developed A3DCP for standardisation protocol [52], which aims to define a general standardisation framework for 3DCP, a digital twin is created via a computational design workflow in Rhino/Grasshopper using Python.

In line with the background research analysis and in the spirit of fostering hybrid standardisation for 3DCP, the proposed digital twin embodies the criteria of performance-based standardisation, a model of flexible standardisation that optimises required functional outcomes without mandating specific technical solutions or materials. In fact, the preliminary materials' characterisation informs the digital twin yet remains disconnected from it.

The goals of the digital twin are as follows:

1. Create a rigorous performance predictive model for compression strength based on experimental data from the preliminary materials' characterisation.
2. Use uncertainty propagation methods to obtain statistical and engineering validation by simulating large numbers of tests.
3. Realise a PoC experiment to test the efficacy of LLMs in interpreting 3DCP multimodal data, including the output from the performance predictive model, and provide targeted adjustment suggestions to maintain 3DPC quality and ensure its final grade.

A Grasshopper Python script, which includes the predictive model equations and the uncertainty propagation, allows for

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TABLE 3 | Adopted research frameworks.

Framework	Role in this study	Key question addressed
Design science research	Overarching structure	How do we systematically create and evaluate a novel artefact (the digital twin) to solve a problem?
Verification and validation	Core methodology/evaluation	How do we build confidence that the digital twin is built correctly and is a credible representation of reality?
Uncertainty quantification	Critical technique within V&V	Given the lack of validation, how can we honestly and quantitatively express the confidence in our predictions?
Case study/proof-of-concept	Research contribution	What new insights or design solutions can we demonstrate using this rigorously built digital twin?

optimising process parameters and maximising 3DPC grade, while the AI-enabled A3DCP simulation uses the optimisation results as a benchmark for process adjustments.

All the work has been conducted in our robotic workshop, equipped with a 3DCP robotic cell featuring a KUKA KR300 R2700-2 industrial robotic arm.

Prompt engineering with LLMs for high-level orchestration and control of CPPS is a spearhead research area that is receiving a significant boost from the latest release of reasoning-optimised models. As mentioned in the background research, relying on specialised AI models is a common approach, with each process task assigned to a different model. The drawbacks in terms of implementation costs have been presented, which define the opportunity for the LLM-based approach.

3.3 | Characterisation of Materials

The implementation of the digital twin starts with the preliminary characterisation of the cementitious material, with uniaxial compression tests always conducted at room temperature after 28 days of curing. Table 4 shows the composition of the adopted commercial ink, a ready-to-use formulation which includes all the necessary elements for 3DCP.

Sixty cubic specimens have been 3D-printed, and their compression strength tested in our lab (Figure 2). A unique ID has been assigned to each specimen, and relevant process parameters have been rigorously monitored and recorded throughout all phases, from material preparation to mechanical testing. In this way, it was possible to establish correlations between the process parameters and the final compression strength and 3DPC grade (Table 5).

The results show that parallel strength is generally lower than perpendicular strength, with an average M15 grade and a population variance of 35, indicating high dispersion. Under these conditions, the anisotropic behaviour together with the low parallel strength poses major structural design and overall performance issues.

3.4 | Digital Twin Model Definition

A digital twin model (Figure 3) of the A3DCP process is created according to the DSR/V&V framework as described hereafter.

3.4.1 | Problem Identification and Motivation

The proposed digital twin must implement a real-time closed-loop control system for A3DCP and ensure scalability. The digital twin and the entire A3DCP model must adapt to various process parameters and setups, thereby promoting standardisation and industrialisation. Adopting a digital twin avoids the development costs of a physical system and the safety issues arising from deploying a self-correcting system with industrial equipment's autonomous behaviour, which can cause damage to the equipment and life-threatening situations for operators.

3.4.2 | Objectives

The digital twin must provide a rigorous model for A3DCP by enabling accurate prediction of material performance, interpreting multimodal data to detect defects or failure modes and suggesting corrective actions in the form of process parameter

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TABLE 4 | Ink formulation per 100 kg.

Component	Mass (kg)	% Of dry powder	Function rationale
Portland cement (OPC)	29.25	29.25%	Long-term strength backbone.
Sulphoaluminate cement (CSA)	15.75	15.75%	Critical: Rapid setting & early strength. Higher % counters low water/PCE delay.
Sand (0–2 mm, well-graded)	52.50	52.50%	Aggregate. Optimised packing reduces water demand. Must be dry & low absorption.
Powdered PCE superplasticizer	1.125	1.125%	Critical: Enables flow at ultra-low water. High-efficiency spray-dried grade.
Powdered cellulose ether (HPMC)	0.035	0.035%	Water retention & cohesion. Use low-viscosity, easily dispersible grade.
Powdered thixotropic agent	0.225	0.225%	Critical: Rapid yield-stress rebuild. Synthetic silica or organics recommended.
PP fibre (6 mm, treated)	0.100	0.100%	Plastic shrinkage control. Must be hydrophilic coated to prevent clumping.

adjustments, actionable equipment commands and corrective/compensatory G-code for manipulators.

3.4.3 | Theoretical Foundations

The digital twin must account for the anisotropic behaviour of 3DPC—caused by interlayer adhesion and the presence of weak planes that affect parallel compression strength, yielding different parallel and perpendicular strengths—and, by optimising process parameters to maximise parallel strength, must prioritise structural safety. Based on the optimised process parameters, the AI will act as a high-level orchestrator in the simulated A3DCP process, detecting defects and anomalies, along with their possible causes, and suggesting real-time adjustments. The bidirectional flow of information between the digital twin and the physical model is completed by incorporating new experimental data into the performance predictive model samples to refine its quality.

3.4.4 | Input Data

The digital twin model must be constructed and optimised based on experimental data and statistical rigour. Due to the inherent variability of the material, an uncertainty-propagation method is adopted to simulate extensive testing and obtain a statistical distribution of mechanical performance. Based on the preliminary materials characterisation, it was possible to identify the relevant process parameters and define the corresponding range of variation. Table 6 shows the parameters included in the digital twin model.

3.4.5 | Model Architecture

The digital twin is implemented in the Rhino/Grasshopper parametric environment, utilising Python coding. The model equations are described in the next section and embedded in the parametric workflow, enabling the optimisation of process parameters to maximise mechanical strength. The Monte Carlo uncertainty propagation method, also part of the parametric workflow, is then used to simulate controlled variations in process parameters and determine the statistical distribution of mechanical performance.

3.4.6 | Validation

The validation of the digital twin, crucial for building credibility in the absence of full-scale experimental correlation, is structured in a pyramid approach, from simple elements to systemic levels. The material model is directly validated against experimental rheological data, providing a physically grounded foundation for the concrete's flow behaviour. The robotic kinematics are considered validated through the high-fidelity Kuka|PRC simulation environment, reproducing our robotic fabrication setup. A notable limitation of using synthetic sensor readings is that it produces idealised, noise-free data that diverges from the stochastic nature of physical sensor readings. Validation of the model's correct formulation is achieved indirectly by ensuring the model accurately simulates the primary governing parameters, such as layer geometry, deposition time and hydration model, which are established in the literature as direct proxies for bond quality and structural stability. At the system level, validation of the digital twin will be completed after comparative physical testing is conducted: This is part of the future developments.

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FIGURE 2 | 3DPC specimen—from 3D printing to compression test.

3.4.7 | Compressive Strength Prediction Model Equations

Based on the knowledge and data obtained from the experimental phase, a compressive strength prediction model was developed, incorporating realistic process-induced variability (e.g., interlayer adhesion, hydration model, curing and environmental conditions), and accounting for 3DPC anisotropy by simulating strength as a bivariate output:

- $f_{c,\perp}$: 3DPC strength with load perpendicular to the layers
- $f_{c,\parallel}$: 3DPC strength with load parallel to the layers

The highest mechanical strength during the preliminary characterisation corresponded to a w/i in the range $0.14 \div 0.15$, yielding characteristic strength values of 19.8 MPa for parallel load and 30.6 MPa for perpendicular load. We defined the operational range starting from the average value 0.145 ± 0.01 (0.135–0.155): In this narrow range, the strength was modelled as a decreasing function of w/i (higher water content ⟶ lower strength), compatibly with cementitious materials behaviour [50] and material characterisation data.

$$f_c = f_{c,\mathrm{ref}} \cdot \left[1 - k \cdot \left(\frac{\mathrm{w}}{\mathrm{i}} - \frac{\mathrm{w}}{\mathrm{i}_{\mathrm{ref}}}\right)\right] (\text{Strength as a function of w/i}), \tag{1}$$

where

- $\mathrm{w/i}_{\mathrm{ref}} = 0.145$, midpoint of $0.135 \div 0.155$
- $f_{c,\mathrm{ref},\parallel} = 19.8$ MPa experimental characteristic parallel strength
- $f_{c,\mathrm{ref},\perp} = 30.6$ MPa experimental characteristic perpendicular strength
- $k \approx 20$, empirical sensitivity: ~20% strength loss per 0.01 increase in w/i

These values are experimental data derived from the material characterisation: By adapting them to the specific properties of the adopted ink and concrete mix, it is possible to scale the digital twin to other production setups.

The perpendicular and parallel compressive strength prediction models are defined as follows:

TABLE 5 | Process parameters and grade of 3DPC specimens (excerpt).

Specimen ID	250612-004	250612-0048	250612-0049	250612-001	250612-005	250612-007
Temperature (°C)				23.9		
Humidity (%RH)				82.8		
Concrete (kg)	99.5	99.5	82.3	99.5	99.5	82.3
Water (L)	14.9	14.9	12.35	14.9	14.9	12.35
Added water (L)				0		
w/i				0.15		
Load orientation	Parallel	Parallel	Parallel	Perpendicular	Perpendicular	Perpendicular
Strength (MPa)	15.7	15.4	16.4	37.0	36.0	30.4
Grade	M15	M15	M15	M35	M35	M30

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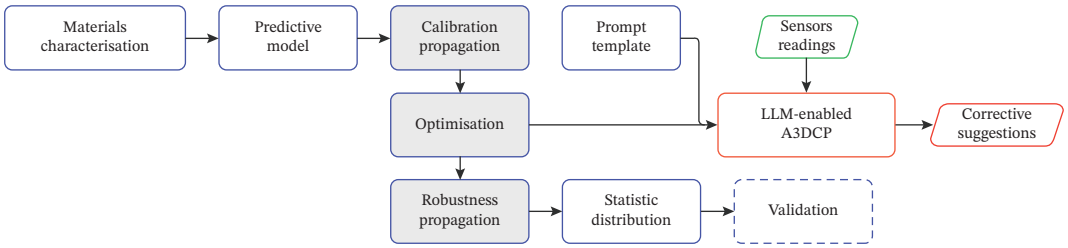


FIGURE 3 | Diagram of the digital twin model.

$$f_{c,\perp} = f_{c,\mathrm{ref},\perp} \cdot S_{\mathrm{w/i}} \cdot S_{\mathrm{cement}} \cdot S_{\mathrm{print},\perp} \cdot S_{\mathrm{env}} \cdot S_{\mathrm{cure}} \cdot \varepsilon_{\perp} (\text{Perpendicular compressive strength}), \tag{2}$$

$$f_{c,\parallel} = f_{c,\mathrm{ref},\parallel} \cdot S_{\mathrm{w/i}} \cdot S_{\mathrm{cement}} \cdot S_{\mathrm{print},\parallel} \cdot S_{\mathrm{env}} \cdot S_{\mathrm{cure}} \cdot \varepsilon_{\parallel} (\text{Parallel compressive strength}) \tag{3}$$

- Water-to-ink sensitivity is as follows:

$$S_{w/i} = 1 - k_w\left(\frac{\mathrm{w}}{\mathrm{i}} - 0.145\right), \quad k_w = 20 (\text{Water to ink sensitivity}), \tag{4}$$

- Cement reactivity factor is as follows:

$$S_{\mathrm{cement}} = 1 + k_{\mathrm{CSA}}(f_{\mathrm{CSA}} - 0.1575), \quad k_{\mathrm{CSA}} = 2.0 (\text{Cement reactivity factor}), \tag{5}$$

(i.e., +1% CSA ⟶ +2% strength)

- Printing factors are as follows:

$$S_{\mathrm{print},\perp} = \left(\frac{12}{h}\right)^{b_1} \cdot \left(\frac{P}{140}\right)^{b_2} \cdot \exp\left(-b_3 (v - 150)^2\right) (\text{Perpendicular printing factor}), \quad b_1 = 0.15, b_2 = 0.15, b_3 = 1.0 \times 10^{-5}. \tag{6}$$

The parallel factor is based on interlayer bond quality and is influenced by

- Layer compaction: better with lower layer height (h), higher extrusion power (P)
- Print speed: too fast ⟶ poor bonding; too slow ⟶ sagging

$$S_{\mathrm{print},\parallel} = \left(\frac{12}{h}\right)^{a_1} \cdot \left(\frac{P}{140}\right)^{a_2} \cdot \exp\left(-a_3 (v - 150)^2\right) (\text{Parallel printing factor}), \quad a_1 = 0.3, a_2 = 0.2, a_3 = 1.5 \times 10^{-5}, \tag{7}$$

- Environmental factor is as follows:

Higher temperature and humidity improve early hydration (but too high RH reduces drying, too low increases evaporation):

$$\begin{aligned} &S_{\mathrm{env}} = (1 + k_T(T - 22.5)) \cdot \left(1 - k_{\mathrm{RH}}(\mathrm{RH} - 60)^2\right) (\text{Environmental factor}), \quad k_T = 0.01 \,\text{per}\,^{\circ}C, (1\% \,\text{per}\,^{\circ}\mathrm{C}), \\ &k_{\mathrm{RH}} = 2.5 \times 10^{-5} \,\text{per}\,^{\circ}\mathrm{C}, (\%\mathrm{RH}) - 2, S_{\mathrm{env}} = (1 + 0.01(T - 22.5)) \cdot \left(1 - 2.5 \times 10^{-5}(\mathrm{RH} - 60)^2\right), \end{aligned} \tag{8}$$

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TABLE 6 | List of parameters used in the digital twin.

Category	Parameter	Description	Nominal	Range	Notes
Material and mix design	w/i	Nominal water-to-ink ratio	0.145	0.135–0.150	Critical for strength; > 0.15 causes sharp drop in strength
	csa_nominal	Nominal cement specific surface area (m^2/g)	15%	10–20	Higher = finer cement = faster reaction
	f_ref_perp	Reference perpendicular strength (MPa)	30.6 MPa	20–40	Measured at 28 days
	f_ref_parallel	Reference parallel strength (MPa)	19.8 MPa	8–25	Always lower due to interlayer weakness
3D printing	h	Nominal layer height (mm)	12 mm	8–14	Thinner layers ⟶ better bonding
	v	Nominal print speed (mm/s)	150 mm/s	80–220	Too fast ⟶ poor interlayer fusion
	P	Nominal extrusion power/pressure (W)	140	110–170	Affects compaction and layer adhesion
	D_n	Nominal nozzle diameter (mm)	30	Fixed	Influences flow and layer geometry
	T_nominal	Ambient temperature (°C)	22.5°C	18–28	Affects hydration kinetics
	RH_nominal	Relative humidity (%)	60%	45–75	Low RH ⟶ surface drying ⟶ weak interface
	t_cure_nominal	Curing time (hours)	678	Fixed	Strength measured at 28 days
	hydration_alpha	Hydration curve shape parameter	0.5	0.2–1.0	Controls early vs late strength gain
	hydration_beta	Hydration rate parameter	0.8	0.1–2.0	Higher = faster reaction

- Curing factor is as follows:

A simplified hydration model is adopted to prioritise real-time optimisation: This choice is supported by the consideration of fixed-age testing at 28 days, making a refined, dynamic hydration model redundant.

Generalised hydration model is as follows [53]:

$$f(t) = f_\infty\left(1 - e^{-\beta t^\alpha}\right). \tag{9}$$

- $f(t)$ Compressive strength at curing time t
- f_∞ Theoretical maximum strength at full hydration ($t \longrightarrow \infty$)
- β Rate constant

  Controls the speed of strength development:

  - Higher β = faster early strength
  - Depends on cement fineness, CSA content, and temperature

- α Avrami exponent [54]

  Controls reaction mechanism shape:

  - $\alpha = 1$: 1D growth (needle-like ettringite)
  - $\alpha = 1.2$–1.5: Typical for blended cements
  - $\alpha = 2$: 2D plate growth
  - $\alpha = 3$–4: 3D spherical growth

  Normalised to nominal time:

$$S_{\text{cure}} = \frac{1 - e^{-\beta t^\alpha}}{1 - e^{-\beta t_{\text{nom}}^\alpha}} (\text{Curing factor}). \tag{10}$$

- S_{cure} Normalised strength gain scales strength relative to a reference curing time
- t_{nom} Reference curing time, baseline age for normalisation (e.g., 168 h = 7 days)

- Residual noise

- $\varepsilon_\perp$: 8%

  Lower variability because perpendicular strength is governed by bulk material properties with less sensitivity to printing defects.

- $\varepsilon_\parallel$: 9%–11%

  Higher variability due to the stochastic nature of interlayer bonding (microvoids, incomplete hydration).

A comprehensive description of the coefficients and magnitudes defined in the equations of the predictive model is presented in Table 7.

This is a comprehensive, production-ready probabilistic material model for the robotic 3DCP digital twin, enabling traceability, statistical rigour and interoperability. The model is based on material-specific coefficients and reference values, making it scalable across different materials after preliminary material characterisation.

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TABLE 7 | List of coefficients and magnitudes defined in the predictive model.

Symbol	Parameter	Value	Units	Range	Direction	Rationale
k_w	w/i sensitivity coefficient	20	% per 0.01 w/i	—	—	For cementitious systems, compressive strength typically follows an inverse relationship with water content. A 0.01 increase in w/c ratio reduces 28-day strength by 15%–25% in conventional concrete [55]. In printable mortars with low w/b ratio, sensitivity is amplified due to: - Critical impact on yield stress and buildability - Effect on interlayer hydration [56] Material characterisation showed strength drop from baseline (w/i ≈ 0.16) to optimised (w/i ≈ 0.14): the corresponding strength increase of ∼15% aligns with $k_w \approx 20$.
k_T	Temperature sensitivity	0.01	% per °C	—	—	Hydration rate approximately doubles per 10°C rise (Arrhenius behaviour). 1% per °C is a conservative estimate for early-age strength development in first 24 h [57]. Lab temperature range (20°C–25°C) yields ±2.5% strength variation.
k_{RH}	Humidity curvature	2.5×10^{-5}	$(\%RH)^{-2}$	—	—	Models how relative humidity (RH) affects 3D-printed concrete strength around an optimal value. Optimal RH ≈ 60% for 3DPC: - < 50%: Surface drying ⟶ weak interlayer transition zone - > 70%: Delayed stiffening ⟶ reduced buildability Quadratic [58] penalty centred at 60% matches experimental observations.
T_{ref}	Reference temperature	22.5	°C	20 ÷ 25	—	Average temperature in the controlled lab range (20°C–25°C)
RH_{ref}	Reference humidity	60	%	50 ÷ 70	—	Average humidity in the controlled lab range (50%–70% RH)
k_{CSA}	CSA sensitivity	2.0	% strength per 1% CSA	—	—	Sulphoaluminate cement (CSA) accelerates early-age strength development via rapid ettringite formation (within 1–4 h) [59]. −1% increase in CSA typically yields 1.5%–2.5% increase in 1-day strength [60]. - Effect diminishes after 7 days but critical for interlayer bonding in 3DPC where layers are deposited within hours.
a_1	Sensitivity of interlayer strength to layer height	0.30	—	—	Parallel	Layer height is a critical parameter for interlayer bonding [61]. Thinner layers increase contact area and penetration depth between layers. Doubling layer height (9 ⟶ 18 mm) typically reduces parallel strength.
b_1	Sensitivity of bulk strength to layer height	0.15	—	—	Perpendicular	Perpendicular strength depends more on bulk material properties than interlayer bonding.
a_2	Interlayer penetration sensitivity	0.20	—	—	Parallel	Higher extrusion pressure forces fresh layer into previous layer, improving mechanical interlock. Critical for bond strength [62].
b_2	Compaction sensitivity	0.15	—	—	Perpendicular	Higher extrusion pressure improves particle packing, but bulk strength is less sensitive than interlayer bonding.
a_3	Bonding time sensitivity (speed decay constant)	1.5×10^{-5}	—	—	Parallel	Narrower optimum (∼130–170 mm/s). Too fast ⟶ insufficient time for interlayer hydration; too slow ⟶ sagging and loss of green strength [56].
b_3	Optimal speed window (speed decay constant)	1.0×10^{-5}	—	—	Perpendicular	Broader optimum (∼120–180 mm/s) because bulk strength is less sensitive to deposition dynamics.

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3.5 | Uncertainty Propagation via Monte Carlo Simulation

A Monte Carlo Simulation is used to simulate thousands of experiments, extracting the mean performance and the 5th percentile (5%ile). Considering the 5%ile means designing for safety [63–65], and applying existing reliability frameworks to 3DPC [66]. The choice to use the 5%ile is as follows:

- Codified in international standards (Eurocode, ISO, fib);
- Rooted in structural reliability theory;
- Essential for safe, compliant design of 3DPC;
- The industry benchmark for material characterisation.

By including uncertainty propagation in the parametric workflow, it was possible to optimise the process parameters and maximise the parallel strength, achieving values above 20 MPa, indicating that the 3DPC grade would be M20 independently of the load direction, thus partially addressing the material's inherent anisotropic structural behaviour.

4 | Results

A Grasshopper parametric workflow based on process parameters, compressive strength prediction model equations and Monte Carlo uncertainty propagation has been implemented. The calibration propagation has been conducted with 1000 Monte Carlo samples to ensure relatively fast computational iterations. A first optimisation was performed with Galapagos in Grasshopper, yielding a single, optimised solution that maximises the 5th percentile parallel strength, i.e., identifying the most robust and reliable printing strategy, thereby aiming at increased safety and reliability rather than just maximising the mean performance. Starting from the Galapagos-optimised solution, a new robustness propagation was conducted with 5000 and 10,000 samples. The process noise, hard-coded in this study but potentially parameterisable, was set to ± 0.002 to reduce scatter, allowing for the simulation of real-world variability within the narrow range of the model's validity, and helping quantify risk by running the simulation thousands of times. All sampled values were logged for further analysis and visualisation (Table 8).

The results are shown in Figure 4 and Table 9. A mean parallel strength of 27.7 MPa and a mean perpendicular strength of 31.8 MPa were obtained. More importantly, the population variation for parallel strength was 10 (71% percentage decrease compared to the materials benchmarking), while the parallel 5% ile was 23.7 MPa and the perpendicular 27.2 MPa (Figure 5), indicating that the 3DPC would achieve grade M20 for both perpendicular and parallel loads (*3DPC* maximum isotropic grade), with relevant consequences for the design of 3D-printed structural elements.

The digital model's normalised perpendicular and parallel strengths were plotted to obtain the normalised cumulative distribution function (CDF). The result is a curve in the range [0, 1] representing the probability of obtaining at least a given strength (Figure 5).

A steep slope of the CDF curve indicates less variability, and within the meaningful range (5%–50%), both curves show steady growth for perpendicular and parallel strengths.

4.1 | AI-Enabled Few-Shot In-Context Closed-Loop Control

Two PoC experiments have been conducted using GLM-4V-plus (mLLM) and GLM-4-plus (LLM), both distributed by Zhipu AI and accessed via a web API: The first experiment aimed at detecting the horizontal shifting of layers and interpreting its nature and reasons [17], and the second at assessing the material quality through CV and superficial analysis [41] and providing corrective instructions. In both cases, ICL was used to limit the nondeterministic, stochastic responses that would complicate the extraction process. The goal of these simulations was to assess

TABLE 8 | Propagation results (excerpt).

w/i	CSA	h (mm)	v (mm/s)	P (W)	T (°C)	RH (%)	t_cure (h)	fiber_vol (%)	str_⊥ (MPa)	str_∥ (MPa)
0.148	0.144	8.478	220.000	150.657	24.028	62.729	168.000	0.180	31.035	27.555
0.145	0.147	8.642	220.000	156.320	23.299	70.842	168.000	0.276	32.455	26.215
0.147	0.139	8.249	220.000	159.989	23.091	68.681	168.000	0.171	26.375	30.142
0.146	0.147	8.000	220.000	161.086	23.417	67.151	168.000	0.174	28.120	26.336
0.148	0.142	8.000	220.000	161.366	23.993	63.670	168.000	0.149	35.098	27.102
0.146	0.141	8.000	220.000	162.885	23.227	68.468	168.000	0.228	32.174	25.630
0.148	0.141	8.000	220.000	166.381	24.481	65.098	168.000	0.272	30.211	24.187
0.145	0.144	8.000	220.000	158.485	23.711	65.068	168.000	0.167	33.750	27.688
0.147	0.145	8.000	220.000	157.380	24.133	63.134	168.000	0.192	29.941	29.580
0.145	0.152	8.314	215.566	164.698	24.710	62.687	168.000	0.291	35.228	24.515
0.146	0.153	8.315	220.000	150.730	24.399	66.516	168.000	0.194	30.575	25.809
0.147	0.149	8.000	220.000	164.691	23.143	70.072	168.000	0.273	31.730	22.635
0.145	0.144	8.606	220.000	155.710	23.209	71.219	168.000	0.270	28.293	25.877
0.147	0.153	8.000	218.948	150.600	24.536	71.987	168.000	0.288	34.243	26.810
0.149	0.146	8.000	220.000	166.525	23.171	68.606	168.000	0.150	27.458	26.223

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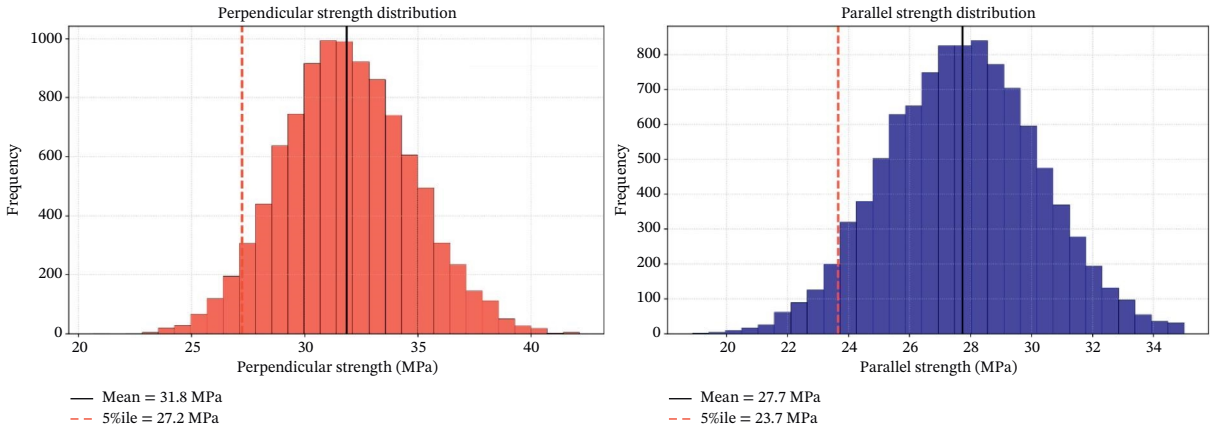


FIGURE 4 | Strength distribution after parallel strength 5%ile maximisation.

the LLM's reasoning quality and reproducibility, and its ability to act as a general orchestrator in an A3DCP process.

A hybrid architecture (Figure 6) comprising specialised models and an LLM as an orchestrator was designed. The specialised models perform low-level analysis, e.g., image recognition and interpretation; the textual output is then passed to the LLM, along with optimised values from the performance predictive model and real-time sensor readings of process parameters, to provide a holistic understanding of the process's current status and determine the LLM's subsequent adjustment suggestions.

The optimised parameters generated by the predictive model and the synthetic sensor readings have been used for the PoC experiments. This was deemed acceptable, as the experiments' aim was to establish the LLM's interpretative capabilities; therefore, no absolute statistical or engineering validation of the predictions was required.

In the first experiment, the mLLM successfully detected and classified horizontal shifting based on profiles extracted from synthetic LiDAR scans (Figure 7).

ICL was used to provide instructions for IS ('carefully focus on the thick black line in the red box') and formatting of the response ('OUTPUT EXAMPLE'), as shown below.

```
# Role: 3D concrete printing analysis expert

## Goals: Precisely detect 3D concrete printing conditions

## Constraints: Be objective, only reply according to the output format, no need to add any other descriptions

## Skills: Familiar with image analysis and following instructions

## Workflow:

After observing the benchmark images, analyse the input image (focus on the thick black line in the red box) and follow the steps below:

### Step 1: Good

Look at the input image and carefully focus on the thick black line in the red box to identify any shifting. If not, conclude it is Good, no need for further observation. If yes, continue to the next step.

### Step 2: Progressive

Look at the input image and carefully focus on the thick black line in the red box to determine whether it bends towards either the left or the right. If yes, conclude it is Progressive. If no, proceed to the next step.

### Step 3: Local

Look at the input image and carefully focus on the thick black line in the red box to identify whether the deviation ever returns to the original axis after a single shift. If yes, conclude it is Local; no further observation needed. If no (e.g. oscillating left-right), proceed to the next step.

### Step 4: Periodic or Random

Look at the input image and carefully focus on the thick black line in the red box to determine whether the shifting resembles sine or cosine waves; if so, conclude it is Periodic. If the oscillation period is variable, conclude it is Random.

## OUTPUT EXAMPLE:
```

TABLE 9 | Perpendicular and parallel grade percentages after parallel strength 5%ile maximisation.

Grade	Perpendicular (%)	Parallel (%)
C16/20	0	0.08
C20/25	0	14.36
C25/30	0.56	68.2
C30/35	21.32	17.2
C35/40	57.2	0.16
C40/45	20.14	0
C45/50	0.78	0

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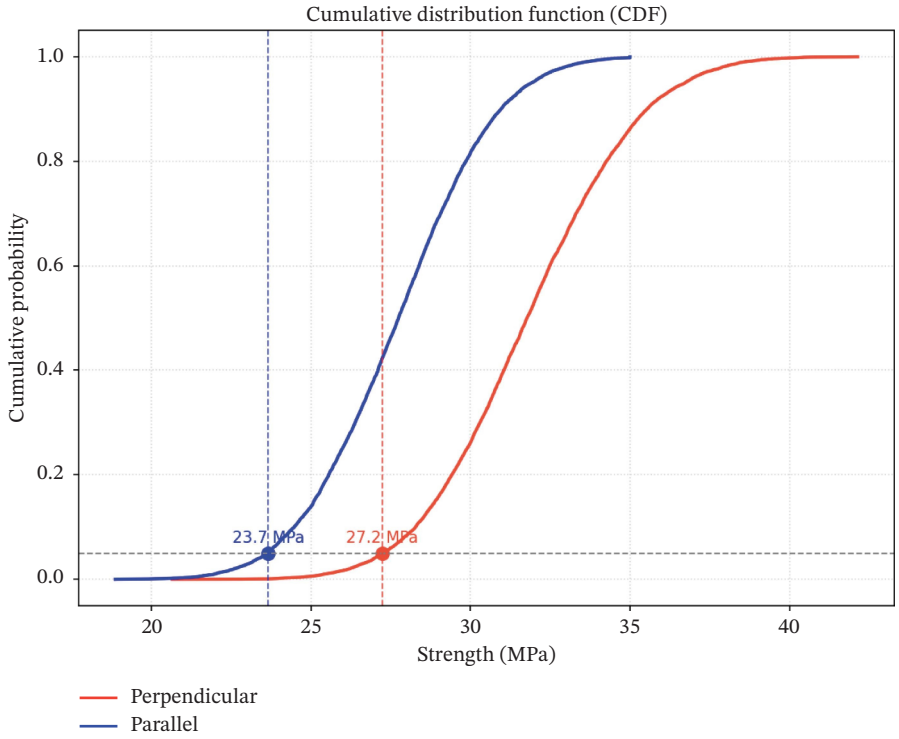


FIGURE 5 | CDF curves showing the 5%ile for perpendicular and parallel strength.

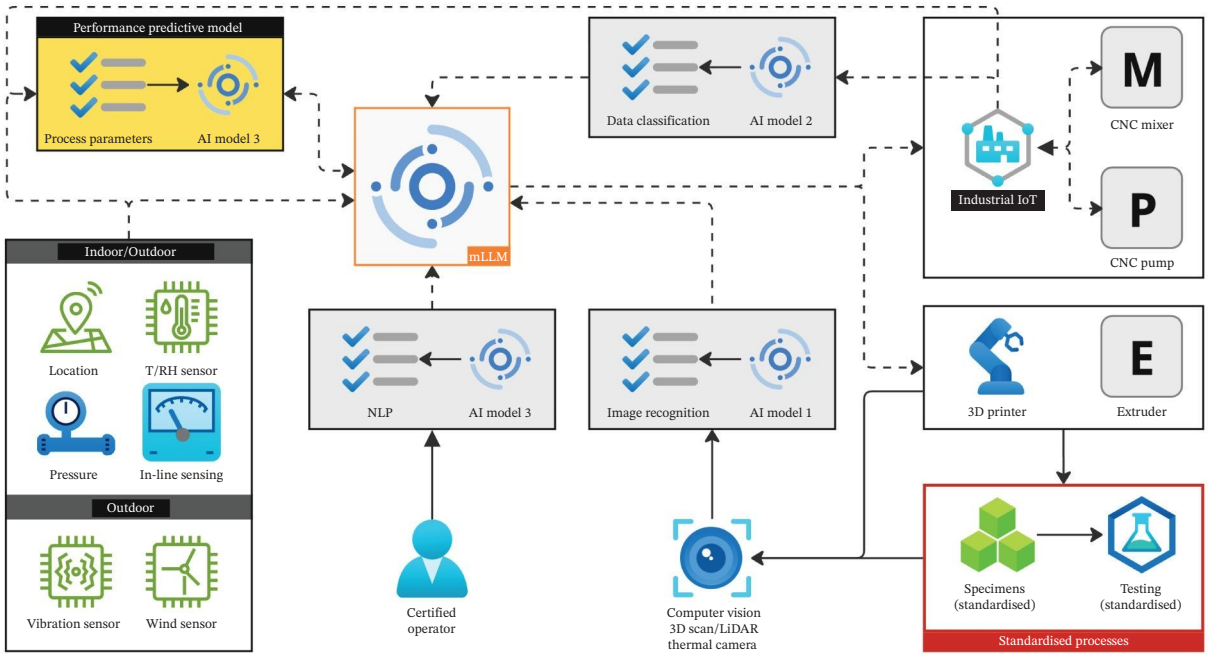


FIGURE 6 | LLM-enabled A3DCP process.

```
Step 1: Yes
Step 2: No
Step 3: No
Step 4: Yes
Category: Periodic
```

In a similar way, in the second experiment, ICL has been used to analyse computer vision images of freshly deposited beads and classify the material hydration as dry, normal or wet (Figure 8).

While dry and normal conditions were easy to detect, the mLLM struggled to interpret wetness, requiring IS to detect localised reflections (groups of bright pixels) caused by superficial moisture.

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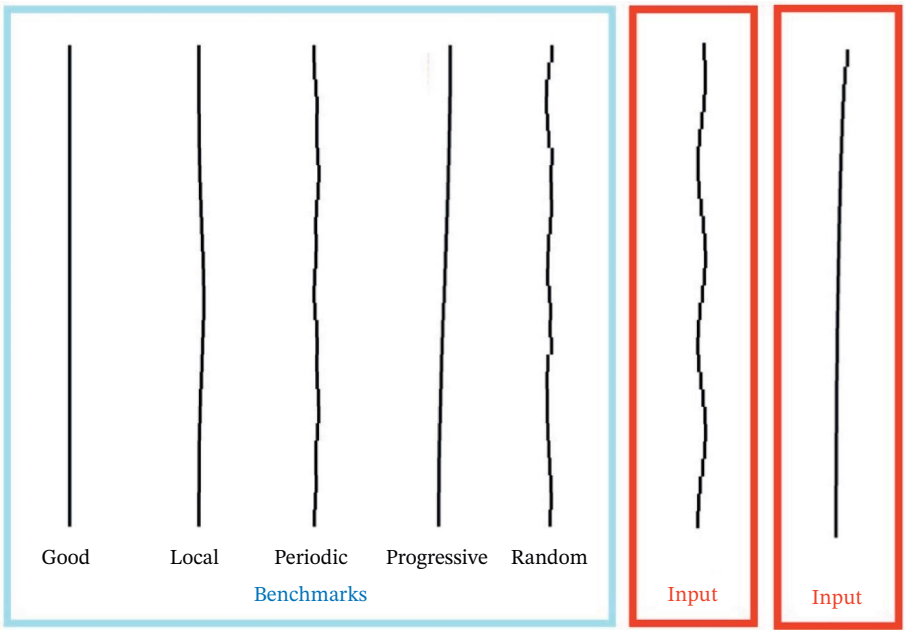


FIGURE 7 | Image segmentation: the benchmark (left) and two different LiDAR sensor readings (right).



FIGURE 8 | Dry, normal and wet 3DPC specimens.

TABLE 10 | Material quality assessment results.

Process parameters	Quality assessment	Suggested adjustments
w/i 0.13 dens. 2300 kg/m^3 T: 15°C RH: 42% Print speed: 100 ∗ 50% h: 15 mm	Dry	Increase w/i Decrease T Increase RH
w/i 0.15 dens. 2250 kg/m^3 T: 13.5°C RH: 44.5% Print speed: 100 ∗ 50% h: 15 mm	Good	No changes
w/i 0.17 dens. 2100 kg/m^3 T: 20°C RH: 70% Print speed: 100 ∗ 50% h: 12.5 mm	Wet	Decrease w/i Increase T Decrease RH

Table 10 shows the simulation results: The extracted suggestions were not translated into low-level, actionable instructions because there was no digital/IoT equipment available.

5 | Discussion

The analyses and results presented in this work aim to address the lack of standardisation for 3DCP materials and processes. The

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proposed approach is that of a hybrid standardisation, with a scalable digital twin tailored to the specificities of our 3DCP cell. Using different inks and equipment might yield different results; however, the model is designed to find the optimal process parameter configuration for any given setup, thereby maximising 3DPC grade.

First and foremost, it is important to acknowledge that, despite the digital twin being developed within a robust DSR/V&V framework, due to the lack of test-bed validation, all results are based on digital simulations and, therefore, on synthetic sensor readings and environmental conditions. The prediction model itself is in its first development iteration (v1.09) and can be improved. All of these aspects constitute criticalities in a real production setting, where sensor noise, model miscalibration and LLMs' latency can lead to localised or global deformations and eventually to structural failure.

3DPC shows good mechanical performance in a narrow rheological window $w/i \approx 0.14$–0.17, slightly broader than the range we have used in our digital twin; therefore, considering a wider range yields relative practical significance. It is reasonable to think that the relationship between strength and w/i can be represented by a smooth monotonic function, be it linear or not. As smooth monotonic functions can be linearised over sufficiently narrow intervals via first-order Taylor approximation, we could assume a linear relationship between strength and w/i in the range 0.135–0.155. This range depends on preliminary benchmarking: Different ink formulations and concrete mixes may yield different ranges within the 3DPC rheological window, thus not compromising the assumption of a linear relationship between strength and w/i. However, if exploring broader ranges (e.g., $w/i = 0.10$–0.20) outside of the typical 3DPC window, it is recommended to switch to a piecewise strength predictive model, considering three different cases:

$$\begin{cases} \dfrac{w}{i} < \dfrac{w}{i_{min}}, & \text{low range,} \\ \dfrac{w}{i} \approx \dfrac{w}{i_{min}} - \dfrac{w}{i_{max}}, & \text{linear window, as defined in this paper,} \\ \dfrac{w}{i} > \dfrac{w}{i_{m}}, & \text{intermediate range,} \end{cases} \tag{11}$$

where:

$$\begin{cases} \dfrac{w}{i_{min}} = \dfrac{w}{i_{av}} - 0.01, & \text{minimum } \dfrac{w}{i}, \\ \dfrac{w}{i_{max}} = \dfrac{w}{i_{av}} + 0.01, & \text{maximum } \dfrac{w}{i}, \\ \dfrac{w}{i_{av}}, & \text{average } \dfrac{w}{i} \text{ (from materials benchmarking).} \end{cases} \tag{12}$$

5.1 | Performance Predictive Model Statistical Validation

Following the DSR/V&V framework, the validation of the digital twin must assess the quality of the performance predictive model and the correspondence between the physical model and the digital twin.

The validation process is articulated in three steps:

1. Prediction quality validation—compare predictions with material characterisation results.
2. Model's physics validation—validation against statistical and engineering metrics.
3. Experimental test-bed validation—new material characterisation and A3DCP.

Prediction quality validation is conducted against the preliminary material characterisation results: verifying that the synthetic data produced by the model aligns with the experimental data indicates the model's correspondence to the physical phenomenon. With reference to w/i, the model's prediction showed a maximised 5%ile parallel grade of M20, corresponding to an optimised value of w/i in the range $0.135 \div 0.150$, confirming the experimental results from the preliminary characterisation campaign where maximum parallel strength was obtained with a w/i ratio of $0.14 \div 0.15$.

The experimental test-bed validation will consist of a new material characterisation campaign informed by the prediction results, thus validating the model's physics and the prediction quality. This experimental campaign, however, can be run only after the predictive model has undergone and obtained full statistical and engineering validation.

5.1.1 | Full Statistical and Engineering Validation

The first version of the computational performance predictive model was characterised by a multiplicative model operating within the same narrow range maintained during the preliminary material characterisation, allowing a linear approximation. Tables 11, 12, 13, 14, and 15 describe the statistical and engineering metrics.

The statistical validation reveals a nuanced performance profile. The model demonstrates excellent calibration of predictive uncertainty—with 90% prediction intervals capturing 89.6% and 89.4% of experimental observations for perpendicular and parallel orientations, respectively—and achieves zero systematic bias in both directions; however, the explanatory power remains modest, as evidenced in Table 16 by coefficient of determination values of $R^2 = 0.129$ (perpendicular) and $R^2 = 0.043$ (parallel).

TABLE 11 | Categorised uncertainty taxonomy.

Uncertainty	Subcategory	Symbol	Description	Perpendicular (%)	Parallel (%)
Aleatory	Irreducible	σ_{ale}	Inherent process variability	30.50	33.48
Epistemic	Parametric	σ_{param}	Calibration uncertainty	30.94	24.50
	Structural	σ_{struct}	Model form inadequacy	34.86	38.27
	Input specification	σ_{input}	Parameter uncertainty	0.00	0.00
Numerical	MC sampling	σ_{num}	Computational error	0.04	0.05
Total epistemic	—		Reducible uncertainty	65.80	62.82

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TABLE 12 | Uncertainty decomposition.

Source	Perpendicular contribution (%)	Parallel contribution (%)	Dominant in
Aleatory (irreducible)	30.50	33.48	Parallel
Parametric (calibration)	30.94	24.50	Perpendicular
Structural (model form)	34.86	38.27	Parallel
Numerical (MC sampling)	0.04	0.05	—
Input specification	0.00	0.00	—
Total explained variance	96.34	96.30	—

TABLE 13 | Core accuracy metrics.

Metric	Symbol	Perpendicular	Parallel	Units
Coefficient of determination	R^2	0.1285	0.0434	—
Adjusted R^2	R^2_{adj}	0.1107	0.0238	—
Root mean square error	RMSE	2.9711	2.6020	MPa
Normalised RMSE	NRMSE	0.1578	0.1639	—
Mean absolute error	MAE	2.4141	2.0974	MPa
Systematic bias	Bias	0.0000	0.0000	MPa

TABLE 14 | Engineering precision metrics.

Metric	Perpendicular	Parallel	Target	Status
Within ±10%	79.0%	70.6%	> 70%	✓
Within ±20%	98.6%	95.2%	> 90%	✓
Normalised error (NRMSE)	15.78%	16.39%	< 20%	✓
Mean absolute deviation	2.41 MPa	2.10 MPa	< 3 MPa	✓
Zero systematic bias	✓	✓	Required	✓

TABLE 15 | Uncertainty quantification metrics.

Metric	Symbol	Perpendicular	Parallel	Target	Assessment
Prediction interval coverage probability	PICP_{90}	89.6%	89.4%	90% ± 5%	Excellent
Mean prediction interval width	MPIW_{90}	9.7749	8.5606	Minimise	Moderate
Interval score ($\alpha = 0.10$)	IS_{90}	11.962	10.508	Minimise	Acceptable
Nominal confidence level	—	90%	90%	—	Fixed
Coverage deviation	ΔPICP	−0.4%	−0.6%	< 5%	Excellent

TABLE 16 | Validation scores.

Validation criterion	Perpendicular score	Parallel score	Weight	Weighted score
Accuracy (R^2)	0.129	0.043	0.25	0.043
Precision (within ±10%)	0.790	0.706	0.20	0.149
Calibration (PICP_{90})	0.896	0.894	0.25	0.224
Bias (zero systematic error)	1.000	1.000	0.15	0.150
Uncertainty sharpness (1/MPIW)	0.102	0.117	0.15	0.016
Overall validation score	—	—	1.00	0.582

The normalised root mean square errors (15.8% and 16.4%) and mean absolute errors (2.41 and 2.10 MPa) indicate acceptable predictive precision for early-stage design applications, with 79.0% and 70.6% of predictions falling within ±10% of experimental values. Uncertainty decomposition reveals that structural model form contributions dominate (34.9% and 38.3%),

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suggesting that the linear polynomial basis inadequately captures the inherently nonlinear physics governing interlayer bonding and anisotropic strength development. Furthermore, sensitivity analysis indicates weak linear correlations between input parameters and parallel strength (all $|r| < 0.10$), implying that key process–property relationships may be obscured by experimental noise or unmodelled interactions. Consequently, whilst the model provides statistically reliable uncertainty bounds suitable for risk-informed decision-making, its limited R^2 underscores the need for either more expressive nonlinear modelling approaches or the incorporation of physics-informed constraints to better represent the complex mechanistic behaviour of 3DPC.

The overall validation score of 0.582/1.000 (Table 16) indicates acceptable performance for early-stage design applications, with excellent calibration ($PICP_{90} \approx 90\%$) as the strongest attribute. Low R^2 values represent the primary limitation and will guide model refinement towards a scalable standardisation tool, as presented in the future developments.

The sensitivity analysis was conducted using Pearson correlation coefficients (r), which quantify linear associations between individual input parameters and the predicted strength. All reported correlations are statistically nonsignificant, reflecting the dominance of nonlinear interactions over simple linear effects. As predicted, the stronger sensitivity of perpendicular strength to w/i ($|r| = 0.244$) aligns with established literature on interlayer bond dependence on paste rheology, while the weak parallel sensitivities suggest that anisotropic strength development is governed by complex, multifactorial mechanisms not captured by linear analysis.

5.1.2 | LLM-Enabled A3DCP Validation

The promising results of the two PoC A3DCP experiments indicate that the LLM can correctly interpret material quality and layers' horizontal shifting. Test-bed validation, quantitative performance benchmarking and the implementation of an optimised design strategy based on the process parameters and performance values obtained from the predictive model will be implemented as future developments to validate these results.

Table 17 shows the advantages of a prompt-engineered LLM architecture over a multiagent architecture.

The simulated AI-enabled A3DCP process used Zhipu AI GLM 4.5, both the LLM and mLLM, for its strong multimodal reasoning capability, although not reasoning-optimised. A reasoning-optimised model would improve ICL and reproducibility. Latency has been disregarded in our experiments as not important in a digitally simulated process: Both the mLLM and the LLM were accessed via a web API, which caused an average 10–15-s lag in the elaboration of the answers. While adopting cloud-based LLMs simplifies hardware setup, it is recommended to deploy local, compressed, fine-tuned versions of the AI models to minimise lag and enable real-time control and adjustments.

During the simulations, the LLM demonstrated its effectiveness in recognising material quality and horizontal shifting patterns, suggesting adjustments to the concrete mixture and a compensatory G-code for the robotic arm. The prompt used for LLM's analysis, interpretation and suggestion tasks was designed to avoid nondeterministic answers and simplify the extraction

TABLE 17 | Comparison between LLM-based and multiagent architecture.

	Prompt-engineered LLMs	**Multiagent**
Training	Not required/light fine-tuning.	Expensive, high-quality fine-tuning.
Implementation	Relatively inexpensive setup with AI-capable hardware and a local LLM.	Each model requires its own setup and dedicated hardware.
Analysis	Prompt engineering – ICL, IS	CNN
Inference	Prompt engineering – ICL, IS	LSTM
Action	Prompt engineering – ReAct	RL policies
Cost (data)	Amortise the data cost across countless tasks via pretraining on a world-knowledge corpus. Its core capability is ICL and instruction-following, allowing it to adapt to new tasks with minimal data via prompts rather than retraining. Leverages vast, pre-trained knowledge for task understanding and decomposition. Requires only few-shot examples or clear procedural instructions via prompting to guide its reasoning for novel tasks, minimising need for task-specific labelled data.	Require concentrated data investment per task. Trained from scratch or fine-tuned on a narrow task. Largely lacks prior knowledge, necessitating large volumes of domain-specific data to achieve robustness. Requires extensive, high-quality labelled datasets *specific to each task* (e.g., thousands of annotated 3DCP crack images). Performance is tightly bound to the quantity and quality of its dedicated training data.

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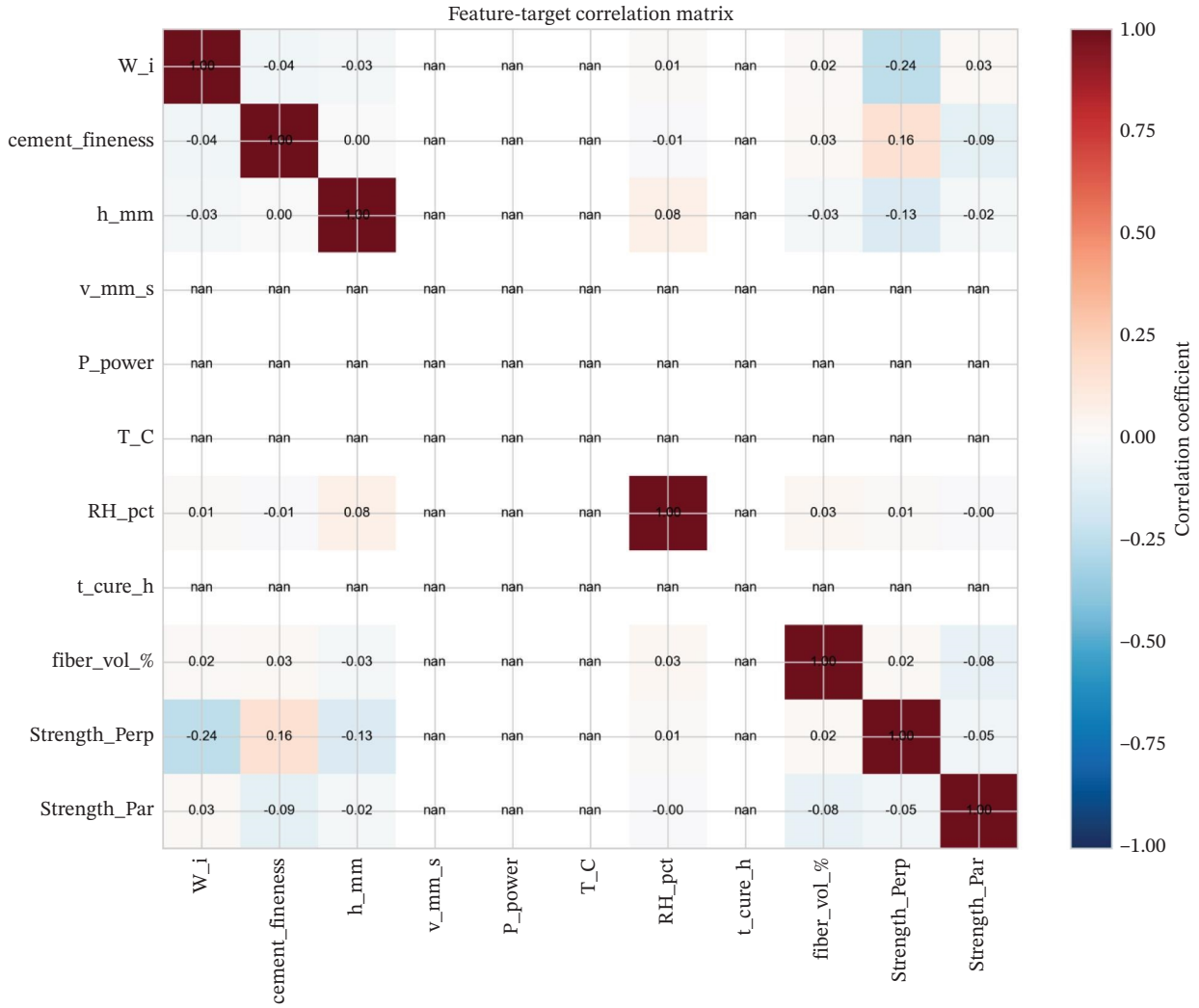


FIGURE 9 | Sensitivity analysis (Pearson correlation coefficients).

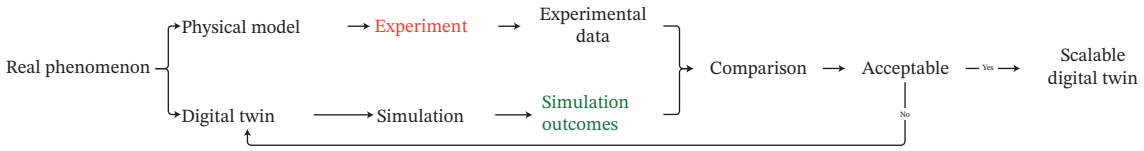


FIGURE 10 | Validation and verification process based on DSR/ASME V&V framework.

TABLE 18 | DoE matrix.

Factor	Low level	High level	Rationale
Water-to-ink ratio	0.140	0.135	Lower w/i = stronger interface
Print speed	65 mm/s	35 mm/s	Better interlayer bonding
Layer height	12 mm	9 mm	Better compaction
Extrusion power	140 W	160 W	Improved bead penetration

phase. It must be noted that the adaptive process requires digital equipment capable of dynamic, automatic adjustments; therefore, a fully IoT-equipped 3DCP lab is required.

Based on the same preliminary materials characterisation data, the LLM's high-level reasoning capabilities were tested on the generic task of identifying potential issues and suggesting possible solutions for interlayer bond quality and reduced parallel strength, focussing precisely on the production parameters identified during the statistical validation (Figure 9). The results obtained from the digital twin after optimisation and propagation are aligned with this LLM's preliminary assessment (parallel

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grade 5%ile M20, 50%ile M25 and 95%ile M30), providing additional qualitative feedback to the quality of the compression-strength predictive model within the adopted narrow range; however, experimental validation is required to fully validate the model.

6 | Conclusions

This exploratory study characterised 3DCP as a complex CPPS, with two main consequences: (1) the need for flexible standardisation; (2) the need for AI as a general controller of the process.

We created a performance-based, scalable digital twin of the A3DCP process, based on experimental materials characterisation and predictive equations for compressive strength, rigorous in the adopted narrow operational range where the linear hypothesis is acceptable, capable of optimising process parameters to achieve the maximum mechanical performance and 3DPC grade, prioritising parallel strength for safety in structural design. We then provided full statistical validation of the performance predictive model.

We highlighted the limitations of specialised AI models, not only because of the considerable effort required for training and fine-tuning but also because of their intrinsic inability to develop a general understanding of the type of problem that may occur in a CPPS process. Therefore, we introduced prompt-engineered LLMs as a high-level orchestrator in the 3DCP process, with their reasoning ability guided via few-shot ICL and other prompt engineering techniques.

We demonstrated the promising potential of prompt-engineered LLMs to act as high-level orchestrators for 3DCP. By interpreting synthetic sensor data and generating actionable commands, LLMs offer a novel pathway for embodied AI, robotic fabrication and industrial automation, potentially reducing the need for specialised AI models and the corresponding financial and technical issues.

We demonstrated how LLMs can analyse, interpret and provide corrective instructions based on sensor readings and predictive model results, showing how the hybrid LLM-based architecture can provide a viable, data-efficient and flexible path to autonomous robotic fabrication control for 3DCP.

We proposed a paradigm shift in A3DCP from 'training-for-control' to 'prompting-for-control', reinforced by recent reasoning-optimised LLMs, with high potential for the AEC industry and for wider fields such as CPPSs and Industry 4.0. The deployment of LLM-orchestrated A3DCP in real construction environments would imply a significant shift in operations and control. In practice, an LLM orchestrator could make the construction process virtually autonomous by implementing a closed loop that translates sensor readings into a synchronised sequence of commands for the manipulator and the material delivery system. This could enhance adaptability, enabling rapid on-site adjustments to unforeseen conditions or design changes.

This study did not account for

- The presence of experimental noise, which is typical of real sensor readings, might affect the LLM's analyses and interpretations.
- The inherent latency of LLMs, which is significant in industrial settings that rely on robotic technologies requiring real-time corrections.

6.1 | Future Developments

In accordance with the adopted hybrid DSR/ASME V&V framework, experimental validation is mandatory, and more precisely, experimental validation of the performance-predictive model; experimental validation of the predicted mechanical strength; deployment of a physical A3DCP system to (1) enable bidirectional information flow within the digital twin, (2) validate the LLM performance during the process, and (3) validate the simulated results (Figure 10).

The first requirement is the experimental validation of the performance predictive model, from which all other aspects of the A3DCP depend; however, given the results of the statistical validation, particularly the low R^2 values, further development of the predictive model is deemed necessary and will be prioritised, so that the experimental validation campaign will start from a more robust basis.

6.1.1 | Updated Performance Predictive Model

The current version of the predictive model is characterised by a narrow operational range and subsequent linear modelisation. While engineering metrics validate the predicted strength, the low coefficient of determination indicates that the model is not scalable, and it can only be adopted in the 3DCP cell where the material characterisation was conducted. Additionally, the sensitivity and correlation analyses indicate that interdependence among process parameters cannot be ignored, suggesting the need for nonlinear elements in the mathematical model.

The new version of the predictive model will adopt more refined hydration and bond models, as well as varying synthetic noise during sample generation (heteroscedastic noise) and nonlinear interaction terms (e.g., interaction between w/i and interlayer bonding), requiring experimental validation across different production settings.

6.1.2 | Experimental Validation of the Predictive Model

Based on the full statistical validation of the predictive model and, in particular, on the sensitivity analysis (Figure 9), it is possible to outline a design of experiment (DoE) matrix by reducing the number of parameters for the new experimental campaign to those having a significant effect on the final 3DPC mechanical performance. Table 18 shows the four most influential parameters related to hydration and interlayer bonding.

The resulting DoE matrix uses a fractional factorial design with only $2^{n-1} = 2^{4-1} = 8$ runs, plus 2 validation runs, while a full factorial design would require $2^4 = 16$ tests. This experimental follow-up will allow us to validate the predictive model's physics and results; however, conducting the validation experiment in the same 3DCP cell used to create the predictive model might not yield a comprehensive assessment. It is, therefore, recommended that the validation experiments be conducted in a different 3DCP cell.

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6.1.3 | Validation of the LLM Response

The qualitative results from the LLM assessment of material quality and horizontal shifting are promising and suggest that an effective prompt engineering strategy can significantly guide and constrain LLMs' nondeterministic, stochastic responses; however, future work with rigorous quantitative benchmarks is required to establish the reliability and validity of these findings.

Validating the LLM's sensing/interpreting capability in a production setting is also essential; in fact, as anticipated in the discussion, the digitally simulated process is not affected by the variable conditions found in production, such as sensor noise and latency. This validation will be fundamental to project the system towards industrial applications. As already discussed, simulating sensor noise patterns and adding a noise filter (e.g., a Kalman filter) to the digital twin can improve the predictive model, while a local deployment of a compressed AI model is essential to avoid variable, web-related latency and rely only on the known model's latency (Table 2). A significant improvement will be the adoption of the new Zhipu AI GLM-Z1-32B reasoning-optimised model, or GLM 4.7. This upgrade will entail better reproducibility, reasoning quality and real-time response (< 100 ms).

Finally, regarding the use of formal grammars to translate natural language outputs into actionable commands, the compensatory G-code extraction for horizontal shifting, which has already been implemented and forms part of another publication currently under review, will be gradually tested.

6.1.4 | Research Directions

This work forms part of an articulated research endeavour under the general research goal of defining a regulatory framework for the standardisation and industrialisation of 3DCP. The three principal research tracks are 3DCP performance-based, decentralised standardisation, A3DCP and stress-informed nonplanar 3DCP. All three tracks are conducted in parallel, and the methods and results of one track inform the others and the general research goal.

The first milestone will be the successful implementation of the A3DCP protocol [52] based on the predictive model presented in this work, with relevant implications not only for process innovation but also for the definition of a standardised process.

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Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

The data, analysis scripts and other materials that support the findings of this study are available from the corresponding author upon reasonable request.

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